

Understanding Voting Difficulties: A Comparison of Democratic and Republican Voters

Datasci 203: Lab 1, Section 001, Group 1

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1 Importance and Context

The question of whether Democratic or Republican voters face greater challenges in exercising their right to vote has become a topic of great importance in American elections. In recent years, voting accessibility and the ease with which citizens can cast their ballots have emerged as critical issues in the United States. With ongoing debates surrounding voter suppression, voter identification laws, and access to polling stations, it is essential to delve into the data and examine the experiences of voters aligned with the two major political parties.

This study addresses the question: *Do Democratic voters or Republican voters experience more difficulty voting?* Understanding the disparities in voting experiences between these two major political groups is not only relevant but also crucial for ensuring a fair and equitable democratic process. By employing statistical analysis of data from the American National Election Studies (ANES), this research aims to show the perceived level of difficulty faced by voters from different political affiliations.

2 Data and Methodology

Our analysis employs the ANES (American National Election Studies) 2022 Pilot Study dataset. This is a cross-sectional survey conducted using non-probability sampling over the internet using the YouGov Panel. This method produces a sample that looks similar to a probability sample on the matched characteristics but may still differ in unknown ways on unmatched characteristics. The YouGov panel is composed of over a million diverse respondents who volunteered to participate in the survey in exchange for points that are redeemable for gift cards. One of the reasons for the administration of this survey was to provide data about voting and public opinion after the 2022 midterm elections in the United States. Respondents were chosen by sample matching to U.S. citizens in the 2019 American Community Survey (ACS) on the basis of gender, age, race, and education ¹.

YouGov sent out a total of 7,437 invitations to participate in this survey. Of the panelists who started the questionnaire, 15 were ineligible non-citizens, 214 did not answer the entire questionnaire (i.e., they are considered breakoffs), and 359 were rejected by YouGov on quality control grounds (such as completion time and item nonresponse). After YouGov conducted initial data cleaning, there were 1585 survey respondents in this dataset.

To tailor this dataset to answer our research question, we first had to merge items that were similar, but used branching logic. For example, YouGov used a branching pipe to ask 50% of survey respondents their political affiliation where the response options began with ‘Republican’ and 50% of respondents where the response options began with ‘Democrat’. After merging the relevant and appropriate items, we then removed respondents who did not identify as either Democrat or Republican (since we are only interested in these two groups). Finally, in operationalizing a ‘voter’, we removed respondents who did not report participating in the most recent election. After these measures, we were left with 817 observations which accounts for 51.5 %

¹ANES. “User’s Guide and Codebook to the ANES 2022 Pilot Study.” (2022).

Table 1: (#tab:- table)Table of count of Democrats and Republicans in the filtered dataset

Party	Number
Democratic	457
Republican	360
Total	817

of the initial responses in the dataset. We also examined the attention check question included in the survey to ensure all remaining data should be used, and we determined that it was not necessary to remove any further respondents from the dataset.

Working from our research question and having already operationalized the concept of ‘voters’ as well as ‘Republicans and Democrats’, we are left to operationalize the concept of ‘voting difficulty experienced’. To do this, we reviewed survey questions asking about factors that would impact survey respondents’ voting experience (e.g., difficulty registering to vote, work schedule, long wait time at the polling place, complexity of voting machine interface, transportation cost to polling place, etc). Because there are a number of ways to operationalize and measure difficulty experienced during voting, we decided to use the self-reported response to the question of “How difficult was it for you to vote?”. Responses were collected using a 5-point Likert scale from 1=Not difficult at all to 5=Extremely difficult. These responses should encapsulate an individual’s overall experience with voting, whereas other questions about wait times or travel times may tend to overweight those experiences as part of the whole experience of voting.

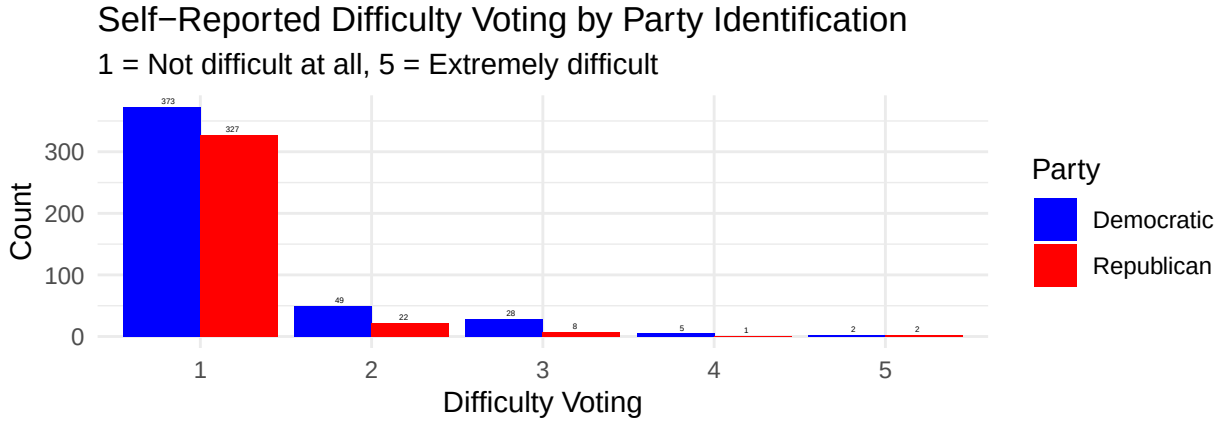


Figure 1: Self-Reported Difficulty Voting by Party Identification

Because of the nature of the variable we use in this analysis, the Wilcoxon Rank Sum (Hypothesis of Comparisons) unpaired test is the most appropriate. The null hypothesis of our test can be formulated as:

Null Hypothesis: *The probability that the difficulty Democratic voters experience is the same as the probability that Republican voters experience.*

To be able to generate valid conclusions, the Wilcoxon Rank Sum (Hypothesis of Comparisons) test requires the following assumptions to be true: (1) the data needs to be ordinal and (2) drawn from an IID (Independent and Identically Distributed) sample. We evaluate each assumption:

1. Data must be ordinal: The survey used a Likert scale to record respondents’ self-reported level of difficulty voting. Answers provided were subjective and categorical, and also ordered.
2. Data needs to be independent and identically distributed: The ANES 2022 Pilot Study surveyed individuals from the YouGov Panel. Some dependencies such as geographical clustering, and similar profiles (e.g., education levels or socio-economic status) may have been introduced because the survey was held online. However, the basis for population inference relies on the assumption of conditional

ignorability (selection on observables). If this assumption holds, the sample distribution of the survey variables will at least approximately match the distribution of those variables in the population. This suggests that these specific dependencies are not likely to occur.

3 Results

```
wilcox.test(small_data$votehard ~ small_data$merged_pid)

##
## Wilcoxon rank sum test with continuity correction
##
## data:  small_data$votehard by small_data$merged_pid
## W = 89937, p-value = 0.0001652
## alternative hypothesis: true location shift is not equal to 0
```

Our analysis, using the Wilcoxon Rank Sum (Hypothesis of Comparisons) test yielded statistically significant results ($W=89937$; p -value: 0.0001652). A p -value below the 0.05 threshold is considered statistically significant. Therefore, we have strong evidence from our test to reject the null hypothesis (Democratic and Republican voters experience the same difficulty voting). Some limitations of our test may affect the conclusions we can derive from this analysis (for example, the ANES survey was conducted using non-probability sampling). Another limiting factor is the ordinal nature of the variable containing Likert scale responses; this type of response limits our ability to describe the data and use appropriate summary measures (e.g., a mean value). Consequently, we decided to use the mode as a measure of central tendency. However, the mode for each group under study (Democrats, and Republicans) was the same (“Not difficult at all”). As a result, we relied on the frequency distribution (Figure 1) and the computed proportion of respondents who experienced any level of difficulty voting. Only 9.2% of Republican voters experienced some level of difficulty voting whereas 18.4% of Democratic voters experienced some type of difficulty voting. Thus, we conclude that for this dataset and the parameters of our research question (e.g., voted in the last election), Democratic voters experience more difficulty voting than Republican voters.

4 Discussion

Even though the ANES survey was conducted using non-probability sampling, we were still able to apply hypothesis testing in the context of statistical analysis by checking possible violations of the assumption of independence and identically distributed (iid) data (e.g., geographical clustering).

When we analyze data, we often look for statistically significant results. This means that the results we see aren’t just happening by random chance. However, just because something is statistically significant doesn’t always mean it is practically important.

For example, imagine we find that Democrats wait, on average, two minutes longer than Republicans at the polls. Even if this result is statistically significant, in real-world terms, a two-minute difference might not matter that much for an individual’s experience.

Based on our dataset, there isn’t a substantial difference in the reported difficulties of voting between Democratic and Republican voters. Each group had respondents who found voting easy and others who found it difficult. However, it is worth noting that some respondents, regardless of their party affiliation, reported that voting was “extremely difficult” for them. This highlights that while party affiliation may not be a significant factor, there are other factors, possibly like location, age, or economic status, that could be influencing the voting experience. These other factors may also be intertwined with political party affiliation at an aggregate level, rather than an individual level (e.g., in geographical areas where there are proportionally more of one party, there may be differences in voting difficulty).

To summarize, while our data suggests there is a significant difference in voting difficulties experienced between Democrats and Republicans, it is important to consider what the statistical difference means in

practice, and it is crucial to ensure that every voter's experience is as smooth as possible.